

One Integer Generates the Discrete Standard Model: the Reading Calculus on D_{IV}^5

Within the type-IV family of bounded symmetric domains, the entire discrete internal skeleton of the Standard Model — its gauge dimensions, chirality, the Majorana neutrino, CP-existence, anomaly-freedom, three generations, the custodial Higgs — is a function of a single measured integer, $n_C = 5$; and the weak force rides on a single trinary question.

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The result (bright-high-schooler version)

Inside the family of shapes our physics lives in, one question sits at the heart of the weak force: is the spinor — the mathematical object a fermion is built from — made on the *real* numbers, the *complex* numbers, or the *quaternions*? That is a single three-way choice, about one and a half bits. The quaternions are the two-sided number system, and that two-ness is exactly the weak force with its paired-up particles (the doublets) — so a “quaternion” answer is what buys the weak force.

Several dimensions n_C give the quaternion answer — so 5 is not the *only* one. What makes it special is that $n_C = 5$ is the smallest shape that passes all four checks at once: the spinor is quaternionic (weak force), the edge is one-sided (a left-handed world), the color piece is real (no hidden color force), and there is room for color to exist at all. Bigger shapes ($n_C = 11, 13, \dots$) pass the same four checks too — and the tiebreaker that says “5, not 11” is a *measured* fact about our world: there are exactly three colors of quark, and the number of colors is just n_C minus two, so three colors means $n_C = 5$.

Everything else in the discrete skeleton of the Standard Model then follows from the same integer: which forces there are and how big, why the world is left-handed, why the neutrino is its own antiparticle, why there are exactly three families of matter, why the quantum anomalies cancel. So the honest headline: the discrete guts of the Standard Model is a function of a single measured integer, $n_C = 5$ —

the *smallest* shape that works, pinned down exactly by the three colors we see.

(A companion result, in Appendix A, steps all the way back to every nicely-curved shape and asks how many independent facts it takes there. The answer is three — the kinds, the sidedness of the edge, the rank. But that is the cross-domain footnote that makes “why these three and no fewer” precise; it is not the physics headline. Within our own family, it is one integer.)

Abstract

We prove that the discrete internal structure of the Standard Model is *generated*, in a precise sense (a declared reading calculus over a declared fact-set), by geometric invariants of the type-IV bounded symmetric domain $D_{IV}^n = SO_0(n,2)/[SO(n)\times SO(2)]$, and that within this family the entire skeleton is a function of the single dimension n_C : the rank is fixed by the family (type IV is rank-2), and the reality-type census and the Shilov orientability are both functions of n_C . The census reduces to a single trinary question — the Frobenius–Schur reality-type of the spinor block — carrying just $\log_2 3 \approx 1.58$ bits of content (the charge block is complex and the color block real for every n in the family; only the spinor moves), and the quaternionic answer is the origin of the weak $SU(2)_L$. $n_C = 5$ is not the unique quaternionic dimension (4, 5, 11, 13, ... all are); it is the smallest domain satisfying four independent, already-proved conditions — quaternionic spinor (weak force), non-orientable boundary (chirality), real color block (no internal color), and $N_c = n - 2 > 1$ (color exists) — and uniqueness among the survivors $\{5, 11, 13, \dots\}$ is the measured $N_c = 3$ (Strong-Uniqueness, K1697). At $n_C = 5$ the reading is the Standard Model: the electroweak gauge dimensions, chirality, the Majorana neutrino, CP-existence, anomaly-freedom, three generations, the custodial Higgs. We give the reading calculus, the assignment table (one row per reading, single-invariant vs conjunction meet), and an honest comparison to Connes’ noncommutative-geometry Standard Model — BST *derives* its finite algebra $\mathbb{C} \oplus \mathbb{H} \oplus \mathbb{R}$ (electroweak, with *no internal color group*) where Connes *inputs* $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ with color gauged; the difference is itself the result. The companion cross-domain minimality (three independent invariants across all bounded symmetric domains, collapsing to one integer within D_{IV}) and the sibling-domain separators are deferred to Appendix A, where they serve the minimality claim; they are not needed for the one-integer lead.

1. The result: one integer, four conditions, the measured tiebreaker

Fix the family to the type-IV domains $D_{IV}^n = SO_0(n,2)/[SO(n)\times SO(2)]$, with dimension parameter $n = n_C$. Three facts determine the discrete internal skeleton — the reality-type census I_1 , the Shilov orientability I_2 , the rank I_3 — and *within D_{IV} all three are functions of n_C* : the rank is 2 for every n (type IV is rank-2; rank-3 fails, T1443), and I_1, I_2 are read off n_C by two periodicities (Section 3). Consequently:

Within D_{IV} , the entire discrete internal Standard-Model skeleton is a function of the single integer n_C , and at the measured value $n_C = 5$

it is the Standard Model.

Why $n_C = 5$ (the honest form — this is Strong-Uniqueness, K1697, already banked, not a new claim). It is *not* that 5 is the unique dimension with a quaternionic spinor: dimensions 4, 5, 11, 13, ... are all quaternionic ($n \equiv 3, 4, 5 \pmod 8$). Rather, $n_C = 5$ is the smallest domain satisfying four independent, already-proved conditions: 1. quaternionic spinor ($n \equiv 3, 4, 5 \pmod 8$) — gives the weak $SU(2)_L$; 2. non-orientable Shilov boundary (n odd) — gives chirality (γ_5 loses global definition, Section 5); 3. real color block ($SO(n-2)$ non-abelian, $n \geq 5$) — gives *no internal color group*; 4. $N_c = n - 2 > 1$ — color exists at all.

The dimensions meeting all four are $\{5, 11, 13, \dots\}$; $n_C = 5$ is the smallest, and the tiebreaker that pins it uniquely is the measured $N_c = 3$ — since $N_c = n - 2$, three colors $\Leftrightarrow n = 5$. So the uniqueness is real but honestly sourced: four *proved structural* conditions cut the field to $\{5, 11, 13, \dots\}$, and one *measured* integer ($N_c = 3$) selects 5. No claim rests on 5 being the only quaternionic dimension — it is not.

The family, checked (every cell computed from Elie's $H_F(n)$, Peirce dimensions (1, $n-2$, 1) — derived, not cited).

n_C	spinor ($n \pmod 8$)	boundary ($n \pmod 2$)	color $SO(n-2)$	$N_c = n-2$	four conditions	verdict
4	$\mathbb{H}$ quaternionic	even $\rightarrow$ orientable	$SO(2)$ abelian $\rightarrow$ complex	2	no (fails 2 and 3)	the checked exception — orientable <i>and</i> complex color (the $SO(2)$ accident)
5	$\mathbb{H}$ quaternionic	odd $\rightarrow$ non-orientable	$SO(3) \rightarrow$ real	3	yes all four	the Standard Model — smallest survivor, and $N_c = 3$
7	$\mathbb{R}$ real	odd $\rightarrow$ non-orientable	$SO(5) \rightarrow$ real	5	no (spinor not $\mathbb{H}$)	foil: real spinor $\rightarrow$ no weak $SU(2)$

n_C	spinor (n mod 8)	boundary (n mod 2)	color SO(n-2)	N_c = n-2	four conditions	verdict
8	$\mathbb{R}$ real	even $\rightarrow$ orientable	SO(6) $\rightarrow$ real	6	no (fails 1 and 2)	foil: real spinor <i>and</i> orientable
11	$\mathbb{H}$ quaternionic	odd $\rightarrow$ non-orientable	SO(9) $\rightarrow$ real	9	yes all four	genuine survivor — excluded only by measured N_c = 3
13	$\mathbb{H}$ quaternionic	odd $\rightarrow$ non-orientable	SO(11) $\rightarrow$ real	11	yes all four	genuine survivor — excluded only by measured N_c = 3

$n = 4$ is the domain least worth discussing and most worth having checked: it is the one place the color block goes complex (SO(2) abelian), and putting it in the table shows the calculus catches its own exception. $n = 11, 13$ are the honest rows — they pass all four *structural* conditions, and nothing structural rules them out; only the measurement $N_c = 3$ does. That is exactly what “one measured integer” means.

The bit count (labeled). The spinor’s reality-type — the one datum that gives the weak force — is a single trinary question, 1.58 bits ($\log_2 3$): the *spinor-block* content. The full family census — the ordered triple (charge, spinor, color) over the distinguishable outcomes it realizes across the family — is 2.00 bits ($\log_2 4$). Either way the content is tiny: the weak sector rides on 1.58 bits, the whole internal census on 2.00. (*The $4.75 = \log_2 3^3$ is the raw capacity of three free blocks — mostly frozen by the family — not the content; 1.58 and 2.00 are the honest figures.*)

(How many independent facts it takes across all bounded symmetric domains — the cross-domain minimality, three invariants collapsing to one integer within D_{IV} — is the companion result of Appendix A, a proof technique for “why these three and no fewer.” The physics lead is the one-integer form above; the siblings appear in the lead only as the checked family table, and in the appendix only as minimality separators.)

2. The reading calculus (declared)

A “reading” is not informal. We fix, once:

- the object — the finite internal space H_F of the domain, its endomorphism algebra $\text{End}_K(H_F)$ under $K = \text{SO}(n) \times \text{SO}(2)$, the Shilov boundary ∂_S , and the domain’s rank;
- the operations R — (i) isotypic decomposition of a representation under a subgroup; (ii) the Frobenius–Schur reality-type ($\mathbb{R}/\mathbb{C}/\mathbb{H}$) of each isotypic block; (iii) generator counting; (iv) the orientability / Pin-cobordism class of ∂_S ; (v) the support-strata count (= rank + 1);
- the fact-set F — the discrete internal observables of the Standard Model (the table of Section 4).

A fact is *generated* if it is the value of an operation in R applied to the object. Minimality is relative to (F, R) : the claim “ k invariants” means *no fewer than k data determine every fact in F under the operations R* . Stating (F, R) is what makes minimality provable rather than rhetorical. (The minimality *proof* — that none of the three can be dropped — is Appendix A; the main text uses the reading calculus to exhibit the skeleton, not to argue minimality.)

3. The invariants, as functions of n_C

- I_1 — the reality-type census of $\text{End}_K(H_F)$: the ordered triple of Frobenius–Schur division algebras ($\mathbb{R}/\mathbb{C}/\mathbb{H}$) commuting with the charge, spinor, and color blocks, multiplicity-free. For D_{IV}^5 it is $(\mathbb{C}, \mathbb{H}, \mathbb{R})$ — a complex charge block, a quaternionic weak block, a real color block. Within the family the charge stays $\mathbb{C}$ and the color stays $\mathbb{R}$; only the spinor type varies, via the spinor reality-type’s Bott/ABS mod-8 periodicity. So the census *content* is one trinary question — $\log_2 3 \approx 1.58$ bits — and the whole weak sector rides on it. (*The $4.75 = \log_2 3^3$ is the raw capacity of three free blocks; most of it is frozen by the family, so 1.58 bits is the honest figure. Multiplicity-free is a hygiene note — it keeps the census a statement about types, not copies; the color dimension $3 = n_C - 2 = N_c$ is a separate rep-dimension reading, never a census multiplicity. The earlier “ $M_3(\mathbb{R})$ ” was the bare-End slip — $\text{End}_{\mathbb{R}}(V_{12})$ ignoring K , whereas the K -equivariant commutant of a real irrep is just $\mathbb{R}$; algebra = $\mathbb{C} \oplus \mathbb{H} \oplus \mathbb{R}$ throughout.)*
- I_2 — the orientability class of the Shilov boundary $\partial_S = (S^{\wedge\{n_C-1\}} \times S^1) / \mathbb{Z}_2$: orientable or not, via the Shilov parity mod 2. For D_{IV}^5 : 5 odd $\rightarrow$ non-orientable (Pin $^-$, $P^2 = -1$). *One bit.*
- I_3 — the rank: fixed = 2 by the family (T1443), so the support-strata count = rank + 1 = 3 generations.

At $n_C = 5$: 5 mod 8 $\rightarrow$ the quaternionic $\text{Sp}(2)$ spinor (the $\mathbb{H}$ block $\rightarrow \text{SU}(2)_L$); $n_C - 2 = 3 \rightarrow$ the real color triple (the $\mathbb{R}$ block, dimension 3 = N_c); 5 odd $\rightarrow$ the non-orientable boundary; rank 2 $\rightarrow$ three generations. The entire skeleton, from one integer — and the one datum that *moves* to give it is the spinor’s trinary type.

4. The assignment table (one row per reading)

The generating set, made explicit (canonical form; single-invariant rows and conjunction “meet” rows; every negative names the invariant it is outside of; the Five-Absence closure block ensures no forbidden particle is left without a row). *The polished figure is the companion assignment grid (Lane C, Grace + Keeper). The separator column names the sibling domain that fails when the invariant is dropped — the full separator argument is Appendix A; here it is a pointer, not the content.*

reading	invariant(s)	separator (Appendix A)	note
gauge dimension = 4 = $\dim(U(1) \times SU(2))$ (<i>uniform census</i> <i>read</i>)	I_1	D_IV ⁷ (census changed)	the three reality-type blocks contribute $\{U(\mathbb{R}), U(\mathbb{C}), U(\mathbb{H})\} = \{0, 1, 3\}$ internal generators — a <i>uniform</i> census read, since $U(\mathbb{R})$ is 0-dimensional — so $0 + 1 + 3 = 4$. No two-line check; both counting rules agree because the $\mathbb{R}$ block sits at multiplicity 1
no internal color group (<i>separate</i> <i>support line</i>)	I_1 + Coleman– Mandula (T265)	D_IV ⁷	two independent arguments, not the count’s second line: (i) the census gives the $\mathbb{R}$ color block 0 internal generators; (ii) its geometric $SO(V_{12})$ rotation is a <i>spacetime</i> symmetry, which by Coleman– Mandula cannot be an internal gauge factor

reading	invariant(s)	separator (Appendix A)	note
anomaly-freedom	I_1	—	self-conjugate $\Leftrightarrow$ nonzero FS indicator $\Rightarrow$ cubic anomaly = 0
custodial SU(2) / $\rho \approx 1$ / no W_R	I_1	—	SO(n) vector $\rightarrow$ (2,2) under SO(n-1)
parity violation (<i>derived</i>)	I_2	D_IV ⁷ vs D_IV ⁸ (orientability pair, K1744)	on a non-orientable boundary the chirality operator γ_5 (a volume-form object) is <i>not</i> <i>globally defined</i> — losing γ_5 is what permits LH $\neq$ RH (T2522)
no sterile neutrino (no ν_R)	I_2	D_IV ⁷ vs D_IV ⁸	the Möbius locus forbids ν_R closure (T1949)
3 generations (no 4th)	I_3	a different-rank domain (D_III ^r)	support-strata = rank + 1 (T2525)
3×3 mixing (CKM/PMNS) structure	I_3	different-rank domain	generation count
chirality (which fermions are LH)	$I_1 \wedge I_2 =$ census_(alg) $\wedge$ twist_(geo)	D_IV ⁷ vs D_IV ⁸ (twist dropped $\rightarrow$ no chirality) and D_IV ⁷ (census changed)	I_2 permits (γ_5 lost on the non-orientable edge); I_1 selects (Y $\neq 0$ makes the rep complex, T2522)
CP violation possible	$I_1 \wedge I_3 =$ census_(alg) $\wedge$ rank_(KM)	D_IV ⁷ (census) and a different-rank domain	the quaternionic $\sqrt{-1}$ (census) <i>and</i> ≥ 3 generations (rank, Kobayashi- Maskawa) — existence only; the magnitude is not read

reading	invariant(s)	separator (Appendix A)	note
neutrino Majorana (<i>not a clean structural meet</i>)	I_2 + one empirical input	D_IV ⁷ vs D_IV ⁸ (twist)	the twist forbids ν_R , so a Dirac mass is unavailable; given the empirical fact that neutrinos are massive, the mass is Majorana. If neutrinos were massless this row is vacuous — so it is twist + one measured fact, not a pure derivation

Meet rows are subscripted algebraic (census) vs geometric (twist) vs count (rank); each meet cites the two separators that fail when its respective invariant is dropped — this is what stops a meet from laundering a value through one invariant. Koide is not in this table — see the next section.

Completeness (these are *all*). The invariants produce exactly the readings above: I_1 ranges over the reality-type of each block (three blocks $\times \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$) and the derived quantities (dimension, self-conjugacy, vector-branching); I_2 is one bit (orientable/not) and its consequences (γ_5 , ν_R); I_3 is the strata count and its square. Every *structural* internal observable is one of these — there is no fourth generator producing a fact off this list (the minimality claim, proved in Appendix A). The Five-Absence set is a consistency check, not a generating row, and it splits by category: *species*-absences (no sterile neutrino, no fourth generation) are already table rows (I_2 , I_3); *process*-absences (no proton decay, no GUT) are dynamical statements — a forbidden *process*, not a missing *species* — and belong to the dynamics (the seam), where they are checked for consistency, not derived here.

4a. What the instrument refuses (lead with Koide) — the strongest evidence it is not an advocacy tool

An honest instrument must refuse things, and the sharpest test is whether it refuses its own best-looking match. It does:

The charged-lepton Koide relation, $Q = (\sum m)/(\sum \sqrt{m})^2 = 2/3$, holds to $\approx 0.0009\%$. It is the tightest numerical coincidence in the whole neighbourhood — and our reading calculus cannot derive it, and we do not claim it here. Koide is a relation among mass *values* (a spectrum fact);

the calculus is *structural*: a fact is readable off I_1 only if it is constant on the reality-type classes $\{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$, and a specific ratio of specific masses is not such a class function. So Koide is provably *outside* the fact-set F . We could have written it as a “census $\wedge$ rank” reading; on the referee’s (and our own) vet, that laundered a value through a structural instrument, and we struck it.

A device that turns away a 0.0009% match because its own stated rules forbid the reading is not an advocacy tool — it is an instrument. This refusal is worth more to a skeptical reader than any positive row, and we put it first among the negatives for that reason. (*Koide is real and may yet be derived — by the mass/eigenvalue instrument, not this one. It sits on the honest ledger as identified-not-derived, elsewhere; disposition: Conditional-Forced, mass sector.*)

5. The chirality mechanism (why the $\mathbb{Z}_2$ is load-bearing)

Here is the mechanism a spin geometer will ask for. Chirality is the eigenvalue of γ_5 , the chirality operator — and γ_5 is a *volume-form* object: it exists globally exactly when the manifold is orientable. On an orientable boundary γ_5 is globally defined, the bulk spinor is vector-like (Witten’s no-go), and there is no room for a chiral theory. On a non-orientable boundary γ_5 is not globally defined — the boundary carries a Pin structure rather than a Spin structure — and *that loss is what opens the chiral sector*: a Pin^- boundary admits a chiral lift where a Spin (orientable) boundary cannot. So the honest statement of the mechanism is T2522: the non-orientability (Pin^-) makes chirality *possible* by removing global γ_5 , and the hypercharge ($Y \neq 0$) *selects* which fermions realize it. Take away I_2 and you lose the handedness (γ_5 becomes global, the theory goes vector-like). (*Scope: Pin-vs-Spin admission is dimension-generic within the admissible types; it is not what selects $n_C = 5$ — the spinor’s trinary type does that. I_2 ’s job is to make chirality possible, not to pin the dimension. The type-level statement — which domains can be non-orientable at all — is the orientability criterion of Appendix A.*)

6. Comparison to Connes’ NCG Standard Model (honest)

datum	Connes NCG	this work	note
the finite algebra	input: $A_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ (chosen to match)	derived: $\text{End}_K(H_F) = \mathbb{C} \oplus \mathbb{H} \oplus \mathbb{R}(I_1,$ multiplicity-free)	<i>the algebras differ, and the difference is the result (below)</i>
three generations	input ($\otimes \mathbb{C}^3$ by hand)	derived (I_3 , rank + 1)	

datum	Connes NCG	this work	note
chirality (γ grading)	input (imposed via KO-dimension)	derived (I_2 , the non-orientable boundary; γ_5 is what the twist provides)	
no ν_R	ν_R typically included	derived (I_2 , Möbius forbids it)	
Higgs potential, gauge couplings	derived (spectral action)	not derived (BST's dynamics are hosted — the seam)	

The difference in the finite algebra *is* the result. Connes takes $A_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ — the $M_3(\mathbb{C})$ is complex, and its unitaries are the internal $SU(3)$: color is an internal gauge group he puts in. BST *derives* $\mathbb{C} \oplus \mathbb{H} \oplus \mathbb{R}$: the color block's K -equivariant commutant comes out real (multiplicity 1, so $U(\mathbb{R})$ is 0-dimensional — no gauge generators), and by Coleman–Mandula the geometric $SO(V_{12})$ rotating the color 3-space is a spacetime symmetry, not an internal gauge factor — so BST derives the electroweak algebra $(\mathbb{C} \oplus \mathbb{H})$ with *no internal color group*, where Connes inputs $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ with color gauged. The gap between Connes' complex $M_3(\mathbb{C})$ and BST's real $\mathbb{R}$ is exactly the census result; we are not reproducing Connes' A_F , we are deriving a *different, smaller* internal algebra and explaining why color is external.

Verdict, stated plainly: BST's novelty is *deriving the inputs* — the electroweak algebra (and that color is *not* internal), the generation count, the handedness, and the missing right-handed neutrino that NCG writes in by hand. Connes' spectral action still derives more of the *dynamics* (the Higgs sector, the unification-scale couplings) than we do, and we do not claim otherwise. The two programs are complementary: one derives the discrete skeleton from geometry, the other the continuous dynamics from a spectral action.

7. Honest scope

The reading calculus is a discrete instrument: a fact is readable off I_1 iff it is constant on the reality-type classes $\{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$. This is *why* the theorem predicts its own negatives — masses, mixing angles, and the CP magnitude are not class functions, so they are provably outside F . They live in a different instrument (Toeplitz/Bergman eigenvalues, the mixing-frame calculus) and are not claimed here. “Structure, not spectrum” is thus a theorem about the calculus, not a hedge. The one dimensional input (a Planck-scale ruler) plus the one measured integer n_C are the honest total cost of the discrete internal skeleton — with the ruler forced by dimensional analysis and n_C forced within D_{IV} by the five-constraint uniqueness (the type-

IV Yang–Mills squeeze). And the internal *content* of that integer, for the census, is one trinary question: 1.58 bits decide the weak force.

Appendix A. Cross-domain minimality (the companion result)

This appendix answers “why these three invariants and no fewer” by stepping back to all bounded symmetric domains. It is a proof technique for the minimality claim, not part of the one-integer physics lead — Casey’s re-centering (Round 17) puts it here deliberately. The sibling domains $D_{IV}^{\{4,7,8\}}$ appear only as separators: foils that fail when an invariant is dropped.

The ladder (three counts, one per reference class). The number of independent generators depends on the reference class; naming the quantifier is what makes each count honest:

rung	quantifier (reference class)	independent generators	what becomes derived
3	$\forall$ bounded symmetric domains D	3: census, orientability, rank	— (all three free)
2	$\forall D$, given Cartan’s classification	2: census, orientability	rank (the census forces type IV, which is rank-2)
1	$\forall D$ in the type-IV family	1: n_C	census + orientability (both functions of n_C)

Going down the ladder is a *trade*: a narrower reference class in exchange for deriving an input. Rung 1 — the physics lead of the main text — is the most specific and cheapest, and the one that applies to the physical domain.

Rung 3 $\rightarrow$ Rung 2 (the census forces type IV). Requiring the finite space to carry a *quaternionic* spinor block (the $\mathbb{H}$ that becomes $SU(2)$), a *real* color block ($\mathbb{R}$), and a *complex* charge circle ($\mathbb{C}$) is met, among the irreducible bounded symmetric domains, only by type IV (the spinor reality-type is a Spin-representation datum selecting the orthogonal family). Once the classification is admitted, the census is the choice of type IV — and type IV is rank-2 — so the rank is derived, not an independent input. Two invariants remain.

Rung 2 $\rightarrow$ Rung 1 (n_C determines census and orientability). Within D_{IV}^n the two remaining invariants are both functions of $n = n_C$: the census through the spinor reality-type (Bott/ABS mod 8) and the color multiplicity ($n_C - 2$); the orientability through the Shilov parity (mod 2). This is the two-clocks structure of the main-text lead.

The siblings as minimality separators (the family check itself is in the lead, Section 1). The six-row family table of Section 1 already shows *that* the siblings differ; here they do the distinct job of proving *non-droppability*. The two invariants are locked to n_C by two different clocks — the census by Bott/ABS mod 8, the orientability by the Shilov parity mod 2 — so neither can be recovered from the other, and a valid separator must move exactly one of them while holding the rest fixed.

K1744 corrects the separator indices. The census separator is D_{IV}^7 (a domain whose census differs from the SM's, so dropping I_1 loses a census-keyed fact). The twist separator is the pair D_{IV}^7 vs D_{IV}^8 — same real-color family, differing Shilov parity (7 odd / non-orientable, 8 even / orientable) — so a fact that flips between them is the fingerprint of I_2 . D_{IV}^4 was never a valid separator: its census is $(\mathbb{C}, \mathbb{H}, \mathbb{C})$, not the SM's $(\mathbb{C}, \mathbb{H}, \mathbb{R})$ — at $n = 4$ the color space has dimension 2 and $SO(2)$ is *abelian*, so the color block acquires a complex structure and goes $\mathbb{C}$. That is an $SO(2)$ -abelian accident at $n = 4$, a *different census*, not a same-census/different-orientability foil. (Earlier drafts cited D_{IV}^4 as the twist separator; it fails because it changes the census too, so it cannot isolate the twist. The valid isolating pair is $D_{IV}^7/{}^8$.)

Elie's orientability criterion (a type-level answer). A Shilov boundary with connected isotropy is orientable; the chirality-enabling non-orientability requires a *disconnected* isotropy (a $\mathbb{Z}_2$ component). Among the irreducible bounded symmetric domains, only types III and IV can carry that $\mathbb{Z}_2$ twist — so most domains cannot host a chiral, Majorana world at all; the requirement is not generic. Positive control: $D_{III}^2 \cong D_{IV}^3$ (the exceptional isomorphism $Sp(4, \mathbb{R}) \cong Spin(3, 2)$) — the one domain in both admissible families; the census and orientability agree across the two descriptions, as they must.

Bit-ceiling. census capacity $\approx 4.75 (\log_2 3^3) + \text{orientability } 1 + \text{rank } (\sim 1-2) < 8$ bits for the whole discrete internal structure across the reference class. *Within* D_{IV} the census content collapses to the single trinary spinor question, 1.58 bits — the figure the main text leads with.

v0.6 — Round 18: lead corrected + locked to Strong-Uniqueness (K1697). “Unique quaternionic” was false (4,5,11,13 quaternionic); the honest form is smallest-of-four-conditions with the measured $N_c=3$ as tiebreaker among $\{5,11,13,\dots\}$. Six-row family table moved into the lead (cells derived from $H_F(n)$); bits labeled (spinor 1.58, family census 2.00). SEP-e split kept ($End_K = \mathbb{C} \oplus \mathbb{H} \oplus \mathbb{R}$; bare $End_{\mathbb{R}}(V_{12}) = M_3(\mathbb{R})$ — not blanket-replaced). Prior: v0.5 — Round 17 re-center (Casey: “why are we even talking about D_{IV}^4 ?”). One-integer / one-trinary-question is the lead; three-invariant cross-domain minimality + all siblings ($D_{IV}^4/{}^7/{}^8$) demoted to Appendix A (companion / proof-technique). K1744 folded: ordered-triple multiplicity-free census (hygiene, not load-bearing); census content = 1.58 bits (not 4.75 capacity); weak force = $n_C=5$ unique quaternionic; twist separator D_{IV}^7 vs D_{IV}^8 ; D_{IV}^4 invalid (census $(\mathbb{C}, \mathbb{H}, \mathbb{C})$, $SO(2)$ -abelian accident); algebra $\mathbb{C} \oplus \mathbb{H} \oplus \mathbb{R}$. Sources: F1060–F1067 (arc); K1724 (census), T1949/T2522 (chirality/twist), T2525 (generations = rank+1), T1443 (type IV

rank-2), T2551 (Connes), T265 (Coleman–Mandula), K1743 (bare-End correction), K1744 (separator/bits correction), Elie’s anomaly + orientability criteria. Ships on Keeper-PASS + Cal-vet + both voices + Casey GO. Nothing pushed; CP existence-only.

— Lyra, Thursday 2026-08-20 (date-verified).